

Nexora Technologies Pvt. Ltd.

SIM-012: The Products We Shouldn't Keep Selling

Case ID	SIM-012
Target Audience	Product Analysts, E-commerce Students & MBA Digital Strategy Students
Primary Skills	Data Cleaning (Product Names, Category Casing), Product Margin Calculation, Return Rate Analysis, Rating Analysis, Marketing ROAS, Product Lifecycle Awareness
Core Competency	Multi-Dimensional Product Portfolio Evaluation; understanding that High Sales does not equal a Good Product
Release Date	August 24, 2026

This package contains the complete case guide: narrative, stakeholder hypotheses, assignment brief, recommended workflow, analytical questions, deliverables, rules, and submission checklist.

1. Executive Briefing & Business Context

Nexora Technologies Pvt. Ltd. is a Bengaluru-based D2C consumer-electronics company selling products directly through its website and mobile application across seven categories: Wireless Audio, Smart Accessories, Mobile Accessories, Smart Home, Computer Accessories, Personal Electronics, and Wearables.

Over the past 18 months, Nexora has aggressively launched new products. The Product team believes a broader catalogue attracts more customers, increases choice, and drives revenue. By all headline metrics, the business appears to be growing: the company now has **500 active products**, has processed **175,522 orders**, and received **69,993 customer reviews**.

However, Finance and Operations are becoming concerned. The growing catalogue appears to contain products that look good on the surface but create hidden problems:

- Some high-volume products have surprisingly thin margins and poor ratings.
- Return rates vary dramatically across products — and returns consume real cash.
- Marketing spend is concentrated on products that may not generate sufficient value.
- Some products are genuinely strong performers receiving too little investment.
- Some products are too new to assess fairly.

"We launched 40 new products this year and revenue went up. But Finance tells me our net margins are falling. I need to know which products are actually worth keeping — and which ones we are just managing for the sake of having a large catalogue." — Head of Product, Nexora Technologies

2. Three Stakeholder Perspectives

Each stakeholder has a different theory about what is happening. These are opinions — your job is to test them with data.

Stakeholder	Their Position	Key Claim
Aarushi Kapoor Head of Product	"The answer is more variety. Our strongest growth has come from new product launches, and our best-selling products prove that customers want choice. We should continue expanding the catalogue."	New products and catalogue breadth drive sustainable growth.
Nikhil Sharma CFO	"We are carrying too many products that look good on the sales report but don't make enough money. Some high-revenue products have weak margins, returns create hidden costs, and slow-moving inventory ties up capital."	Revenue is hiding poor product-level economics.
Sonal Verma Head of Marketing	"Some products are not failing — they are simply under-promoted. With adequate marketing exposure, products that currently appear weak would show their real potential."	Low sales can be explained by low marketing investment, not poor product quality.

3. Available Datasets Overview

You have been provided with seven relational datasets:
products.csv (500 products with name and category naming anomalies), sales_orders.csv (~175,000 orders), customers.csv (50,000 customers), marketing_campaigns.csv (200 campaigns), product_campaign_performance.csv (~4,300 rows), product_reviews.csv (~70,000 reviews), and returns.csv (~20,000 returns with some missing return reasons).

Your Assignment

Your Role

You are a **Junior Product Analyst** at Nexora Technologies. The Head of Product has commissioned an independent product portfolio review. You have access to all sales, customer, campaign, review, and returns data.

Your Central Mission

Evaluate Nexora's product portfolio and determine which products are creating genuine business value, which have hidden weaknesses, and where management resources should be allocated. Present a defensible, evidence-based product recommendation.

IMPORTANT:

You must evaluate each product across multiple dimensions. A product with high sales is not automatically good, and a product with low sales is not automatically bad. Context — including product age — matters.

Do not recommend dropping recently launched products based on limited data. A strong analyst knows when to say: 'We don't have enough evidence yet.'

Your Investigation Must Address These Business Questions

- 1. Revenue vs. Profitability:** Does higher sales volume translate into higher margin contribution? Which products generate strong net revenue and which rely heavily on discounts to drive volume?
- 2. Return Rates & Hidden Costs:** Which products have unusually high return rates? What are the dominant return reasons? How does the refund cost affect realized value?
- 3. Customer Feedback:** Do customer ratings align with sales performance? Which products receive strong reviews despite low sales? Which receive poor reviews despite high sales?
- 4. Marketing Investment vs. Returns:** Which products receive the most marketing spend? Is that spend generating proportionate attributed revenue (ROAS)? Are any low-sales products simply under-promoted?
- 5. Product Age & Fairness:** When did each product launch? How does product maturity affect the interpretation of sales volume and review count? Which products are too new to evaluate confidently?

Recommended Analytical Workflow

Phase	Activity	Description & Deliverable
01 — Understand	Read & Frame	Identify the central question and plan your investigation across cost, returns, ratings, and marketing dimensions.
02 — Inspect	Schema Review	Map all seven tables. Understand the relationships between products, orders, reviews, returns, and campaigns.
03 — Clean	Data Preparation	Standardize product names (casing, trailing spaces), unify category naming inconsistencies. Document your resolutions.
04 — Baseline	Portfolio Overview	Calculate order volume, gross sales, net sales, discount rate, and gross margin per product. Rank by net sales and by margin.
05 — Returns	Return Rate Analysis	Calculate return rate (returns / orders) and total refund cost per product. Identify dominant return reasons.
06 — Ratings	Customer Feedback	Calculate average rating and total review count per product. Identify rating category concentrations for poor-rated products.

07 — Marketing	Campaign Efficiency	Calculate total campaign spend and ROAS (attributed revenue / campaign spend) per product. Identify products with high spend but poor ROAS.
08 — Age	Product Maturity	Calculate product age (days since launch_date). Flag products launched within the last 90 days as 'Immature' — exclude from definitive judgments.
09 — Integrate	Multi-Dim Analysis	Combine margin, return rate, rating, and ROAS into a per-product evidence summary. Identify archetypes: Stars, Revenue Traps, Hidden Opportunities, Weak Products.
10 — Conclude	Evaluate Hypotheses	Test all three stakeholder positions with data. Determine which are supported, partially supported, or contradicted.
11 — Recommend	Actions	Recommend per-product actions: Invest / Improve (Marketing) / Improve (Product) / Maintain / Reconsider. Back each with evidence.

What You Are Expected to Investigate

A. Product Revenue & Margin Baseline

- Calculate gross sales, net sales (after discounts), and gross margin per product.
- Calculate average discount percentage per product. Which products rely on heavy discounting to drive volume?
- Identify the top-10 products by gross sales and compare their rankings by gross margin. Are they the same list?
- Which product categories show the strongest and weakest average margins?

B. Return Rate & Refund Cost Analysis

- Calculate return rate (returned orders / total orders) per product.
- Calculate total refund value per product. Which products are costing Nexora the most in refunds?
- Analyse return reasons by product. Are returns driven by product defects, quality issues, or customer expectations?
- Note: Some return records have missing return reasons. Document how you handle these gaps.

What You Are Expected to Investigate (Cont.)

C. Customer Ratings & Review Analysis

- Calculate average customer rating per product.
- Compare the rating profile of high-sales products vs. low-sales products. Do customers agree with the sales rankings?
- Identify the dominant review categories for low-rated products. What dimension are customers complaining about?
- Note: Not every customer leaves a review. A low review count does not mean a bad product.

D. Marketing Investment & Campaign Efficiency

- Calculate total marketing spend and ROAS (attributed revenue / campaign spend) per product.
- Which products receive the highest marketing investment? Are they generating proportionate returns?
- Which products receive very little campaign exposure? Could their low sales be explained by marketing neglect rather than poor product quality?
- Evaluate the Marketing Head's hypothesis: does higher campaign spend reliably lead to better performance?

E. Product Age & Evaluation Fairness

- Calculate days since launch for each product using the launch_date field and the dataset end date.
- Flag products launched within the last 90 days as 'Immature'. How many orders and reviews do these products have?
- Explain in your presentation why immature products should not be judged by the same standards as established products.
- For immature products, recommend a timeline for re-evaluation rather than a definitive verdict.

KEY GUARDRAIL:

Do not use a single metric to classify products. Evaluate each product across margin, return rate, customer rating, marketing efficiency, and product age before reaching a conclusion. Different dimensions may tell different stories about the same product.

What You Must Submit

Deliverable 1 — Executive Presentation (10–12 Slides)

1. **Executive Summary:** 3–4 key portfolio findings.
2. **Business Context:** Why high sales does not equal a good product.
3. **Data Preparation Log:** Product name anomalies, category inconsistencies, missing return reasons.
4. **Portfolio Baseline:** Orders, gross sales, net sales, discount rates, margins by product.
5. **Return Rate Analysis:** Return rate by product, dominant reasons, refund cost.
6. **Customer Ratings:** Average ratings, review category profiles.
7. **Marketing Efficiency:** Campaign spend, ROAS, exposure gaps.
8. **Product Age:** Maturity classification, immature product handling.
9. **Product Portfolio Map:** Multi-dimensional evidence summary per product.
10. **Hypothesis Evaluation:** Verdicts on all three stakeholder claims.
11. **Prioritised Recommendations:** Invest / Improve / Maintain / Reconsider per product category.
12. **Limitations:** What additional data would improve your confidence.

Deliverable 2 — Analytical Workbook

A formatted Excel workbook with cleaned product data, pivot tables for margin, return rate, ratings, and campaign ROAS analysis, and a product summary dashboard. All calculations must use formulas.

Deliverable 3 — SQL Script File

A well-commented .sql file with sections: 01_Data_Cleaning, 02_Sales_Margin_Baseline, 03_Return_Analysis, 04_Rating_Analysis, 05_Campaign_Efficiency, 06_Product_Age, 07_Portfolio_Summary.

What Good Analysis Looks Like

■ WEAK:

"PowerBank 20K has the most orders (85,000). AirBeats Pro is also popular. CableKit Pro and SmartTracker Tag sell less and should be discontinued."

- Reads a single metric (orders). Ignores margin, return rates, ratings, and marketing. Wrongly penalises the SmartTracker for low sales without checking why.

■ ■ BETTER:

"AirBeats Pro (PRD-001) has 4.7/5 rating and 2.0%% return rate, while PowerBank 20K (PRD-002) has 3.2/5 and 18.1%% return rate. CableKit Pro (PRD-004) has a 2.8/5 rating and 22.5%% return rate. SmartTracker Tag (PRD-003) has only 7,000 orders."

- Correctly identifies dimension differences but does not explain SmartTracker's low sales, does not calculate effective margin, and does not test the marketing hypothesis.

■ STRONG:

"Nexora's portfolio contains at least four distinct product situations. AirBeats Pro (PRD-001) is a genuine star: 45%% gross margin, 2.0%% return rate, 4.66/5 rating, and 9.5x ROAS — every dimension confirms investment is warranted. PowerBank 20K (PRD-002) is a revenue trap: 85,000 orders, but 8%% gross margin, 18.1%% return rate, 3.2/5 rating, and 1.53x ROAS. Discount-driven volume is masking a deeply unprofitable product. SmartTracker Tag (PRD-003) is a hidden opportunity: 52%% gross margin, 3.0%% return rate, 4.56/5 rating — but only INR 40,000 in campaign spend. Its low sales are explained by marketing neglect, not product weakness — supporting Sonal Verma's hypothesis for this specific product. CableKit Pro (PRD-004) is a weak product: 8%% margin, 22.5%% return rate, 2.8/5 rating, and 0.53x ROAS — no dimension supports continued investment. SoundBar Mini (PRD-005) is too new to judge — 6 weeks of data is insufficient. A 90-day re-evaluation is recommended."

- Integrates all five dimensions; names and explains each archetype; challenges the Procurement Head's catalogue-expansion hypothesis; validates Marketing Head's claim for one specific product while rejecting it for another.

Rules of the Investigation

1. **Evaluate across multiple dimensions:** Revenue + Margin + Return Rate + Customer Rating + Marketing Efficiency + Product Age.
2. **Clean before you measure:** Standardize product names and category names before running any aggregations.
3. **Do not use a single metric:** High sales alone does not make a product worth keeping. Low sales alone does not make it worth dropping.
4. **Respect product age:** Exclude products launched within the last 90 days from definitive judgments.

5. **Handle missing data explicitly:** Document how you treat missing return reasons and products with no reviews.
6. **Distinguish discount-driven sales:** Check whether high order volumes are supported by adequate margins.
7. **Test all three hypotheses:** Evaluate the Product Head, CFO, and Marketing Head claims with evidence.

Evaluation Criteria & Weights

Criteria	Weight	What Graders Look For
Data Quality & Preparation	10%	Product name cleaning, category unification, missing return reason handling.
Analytical Methodology	20%	Correct margin calc, return rate formula, ROAS, product age calculation.
Business Investigation	25%	Testing all three hypotheses; identifying all five product archetypes with evidence.
Technical Accuracy	15%	Correct SQL joins (LEFT JOIN for reviews/returns), NULL handling, date functions.
Business Insights & Reasoning	15%	Distinguishing revenue traps from stars; naming hidden opportunities; acknowledging immature products.
Recommendations	10%	Product-specific Invest/Improve/Maintain/Reconsider recommendations with evidence.
Communication	5%	Readable slides, clean Excel, all numbers verified across SQL, Excel and slides.

Submission Checklist

Check	Task Description	Status
[]	Standardized product names (casing, trailing spaces) and category inconsistencies.	Confirmed
[]	Documented treatment of missing return_reason records.	Confirmed
[]	Calculated gross margin % per product (selling_price - unit_cost) / selling_price.	Confirmed
[]	Calculated net sales per product after discounts.	Confirmed
[]	Identified and noted heavily discounted products (avg discount > 25%).	Confirmed
[]	Calculated return rate (returns / orders) per product.	Confirmed
[]	Identified dominant return reason categories for high-return products.	Confirmed
[]	Calculated average customer rating per product.	Confirmed
[]	Calculated total campaign spend and ROAS per product.	Confirmed
[]	Flagged products launched within 90 days as Immature with withhold-judgment verdict.	Confirmed
[]	Evaluated all three stakeholder hypotheses (Product Head, CFO, Marketing Head).	Confirmed
[]	Presented a product-level recommendation for at least 5 representative products.	Confirmed
[]	SQL script organized in 7 labeled sections, all queries commented.	Confirmed
[]	Presentation has 10-12 slides, all numbers verified against SQL and Excel.	Confirmed